Review retrospective:
the bounded-prior test-function primitives — an honestly-in-progress file, two sound lemmas, three anticipatory deps removed

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Abstract

A soundness review of `Laplace/OneD/QuarticBoundedPriorTestFn.lean`: the two foundational primitives (partition positivity and an inner-shell lower bound) for the still-pending bounded-prior localisation $\int_{-a}^{a} \varphi \, e^{-tw^4/24} \sim \varphi(0) Z_a(t)$. Result: **a clean fidelity pass**, three anticipatory dependencies removed (a dead import and two dead opens), and one docstring hypothesis disambiguated — with the key judgement being that the file's "(in progress)" status is *accurate* and must not be "fixed."

1 Setting

The φ-weighted companion to QuarticBoundedPrior, and the last small never-reviewed laplace `OneD` file. Its eventual headline is the localisation $\int_{-a}^{a} \varphi(w) e^{-tw^4/24} \, dw \sim \varphi(0) \, Z_a(t)$ $(t \to \infty)$, packaged as an `IsLittleO` at `atTop` — but that has not been formalised yet. The file currently holds only the two building blocks.

2 What was reviewed

`quartic_bounded_prior_partition_pos` ($Z_a(t) > 0$ for $a > 0$, any t) and `quartic_bounded_prior_partition_lower_bound_inner` ($2\delta \, e^{-t\delta^4/24} \le Z_a(t)$ for $0 < \delta \le a$, $0 \le t$). The informal ground truth was the file's docstrings and the elementary Gaussian-type estimates they describe.

3 Fidelity / soundness findings

Sound; no defect in either statement. Positivity is the strictly-positive integrand on a nonempty interval (support is all of $\mathbb{R}$, $\mathrm{vol}([-a, a]) > 0$). The lower bound restricts to the inner shell $[-\delta, \delta]$, where $w^4 \le \delta^4$ and $t \ge 0$ give $e^{-tw^4/24} \ge e^{-t\delta^4/24}$; `setIntegral_const` contributes the length 2δ, and subset monotonicity lifts $\int_{[-\delta,\delta]} \le \int_{[-a,a]}$. The reviewer confirmed both statements — constant 2δ, exponent $t\delta^4/24$, hypotheses $0 < \delta \le a$, $0 \le t$, inequality direction.

The crucial non-finding: the "in progress" status is true. The module docstring says the file is in progress and the headline "will land in a follow-up session." The immediately-preceding HarmonicGibbsObservableMonomials review caught exactly this phrasing being *stale* (the headline had in fact landed in-file). So this one was checked the same way — and came out the other way: a seabed grep confirmed nothing imports this file, nothing consumes its two lemmas, and the localisation headline exists nowhere (the seabed's `Localisation.lean` is the *harmonic* localisation, a different result). The status is faithful and was left untouched. Only one genuinely imprecise docstring bullet was fixed: the Status line wrote "$Z_a(t) > 0$ for $0 < a, t : \mathbb{R}$," which misreads as requiring $0 < t$, whereas the theorem needs only $0 < a$ (the per-theorem docstring already said so) — disambiguated to "$0 < a$ (any t)."

4 Simplifications applied

Three anticipatory dependencies, all dead now. The import `Mathlib.Analysis.Asymptotics.Defs` was added for the future `IsLittleO` headline but is unused by the two primitives (the only `=o[atTop]` is in the docstring); `open Filter` and `open Topology` were likewise anticipatory (all `Filter.*` names are qualified, no $\mathcal{N}$/`nhds` appears). The one trap checked carefully — whether the $\leq^{\mathrm{m}}$ $[\mu]$ a.e. notation needs `open Filter` — resolved against it: the notation is available without it. `open MeasureTheory` and `open Set` stay. Build green, no in-file warnings, signatures byte-identical; the reviewer affirmed.

5 Disagreements and open questions

None. The reviewer noted the routine caveat that removing an import narrows the transitive surface for any external downstream relying on it indirectly — moot here, since nothing imports this file.

6 Lessons

- **"In progress" is a claim to verify, not a reflex to fix.** Two consecutive reviews hit the identical "will land in a follow-up" phrasing; one was stale (fix it), the other accurate (leave it). The discriminator is a thirty-second seabed grep — does the headline exist anywhere? — not the phrasing itself. Treating "in progress" as automatically-stale would have introduced a false docstring here.
- **Anticipatory imports/opens accrete on in-progress files and are safe to shed.** A file written headline-first often opens `Filter`/`Topology` and imports `Asymptotics` for a theorem that has not arrived; until it does, they are pure dead weight, and the build confirms their removal. (They can always be re-added with the headline.)

7 Follow-ups

- The genuine forward tide this file is waiting on: the bounded-prior localisation headline $\int_{-a}^{a} \varphi\, e^{-tw^4/24} \sim \varphi(0) Z_a(t)$ (`IsLittleO` form), for which these two primitives are the building blocks.
- The last never-reviewed laplace `OneD` file is the large `IntegralRemainder`.